

Manitoba Pre-Calculus 40S Exam

1. (3 points) Gina correctly started to answer the following question. Complete her solution.

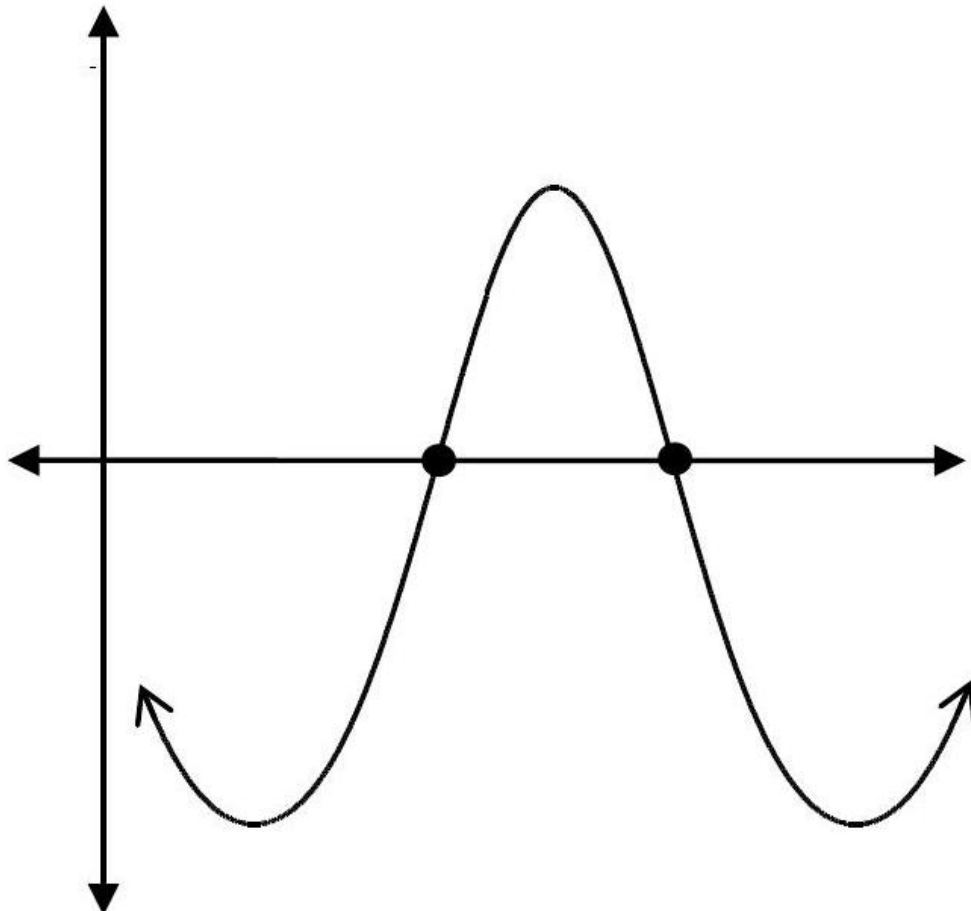
Question: Solve the following equation for all real values of θ .

Express your answer in radians correct to 3 decimal places.

$$3 \sin^2 \theta - 14 \sin \theta - 5 = 0$$

Gina's solution: $3 \sin^2 \theta - 14 \sin \theta - 5 = 0$

$$(3 \sin \theta + 1)(\sin \theta - 5) = 0$$



2. (3 points) Find and simplify the 6 th term in the binomial expansion of $(3x^4 - \frac{1}{x^3})^9$.
3. (3 points) The number of times a website is visited can be modeled by the function:

$$A = 800(e)^{rt}$$

where A = the total number of visitors at time t

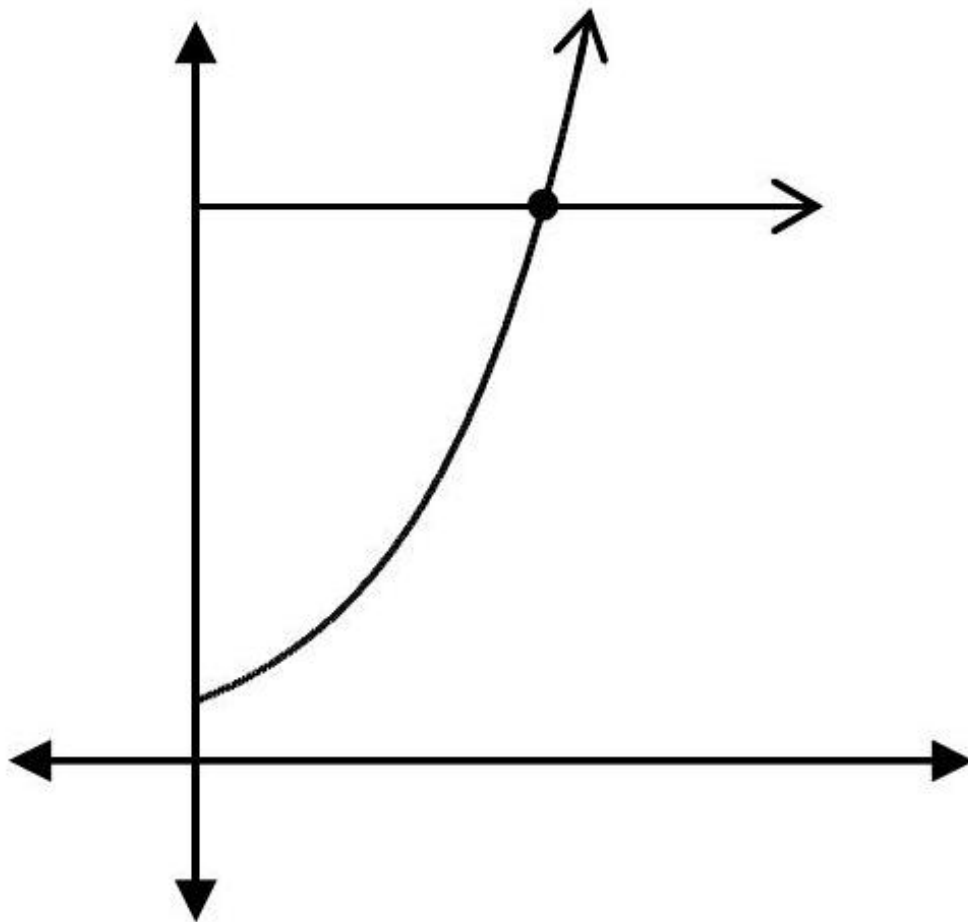
t = the time in days ($t \geq 0$)

r = the rate of growth

After 5 days, 40000 people have visited the site.

Determine the number of visitors expected after 9 days.

Express your answer as a whole number.

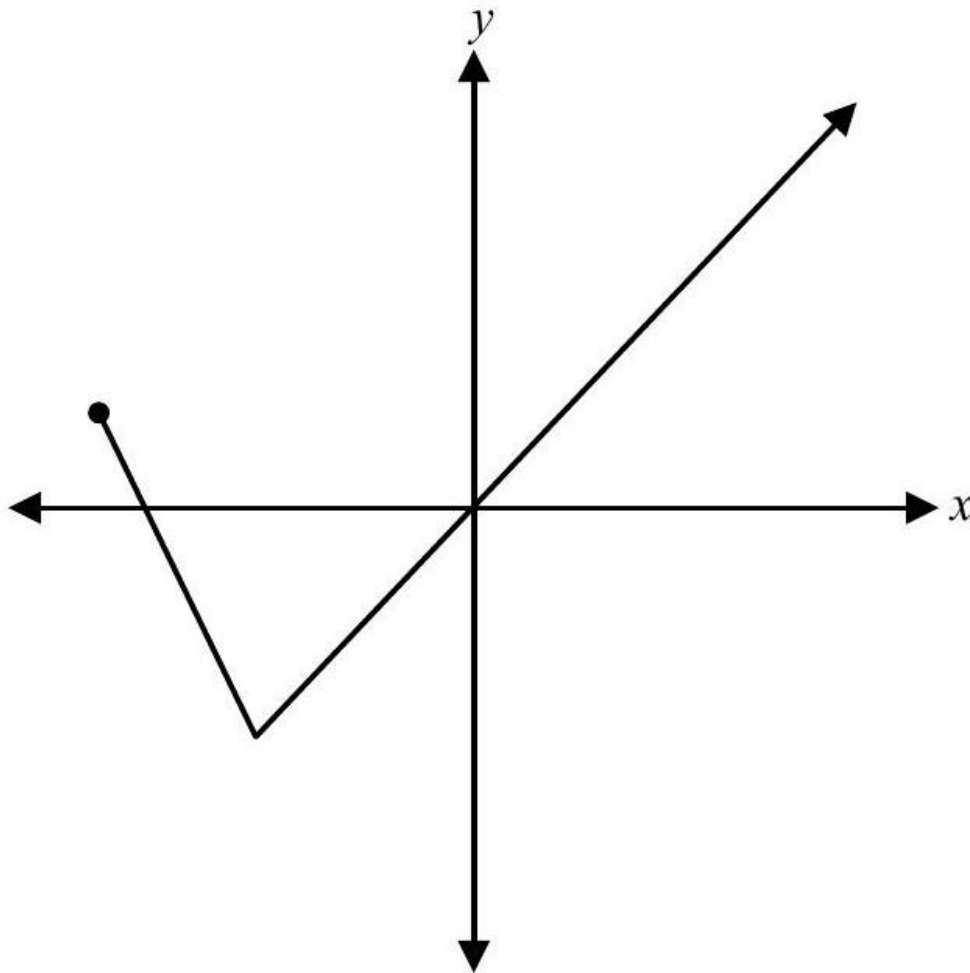


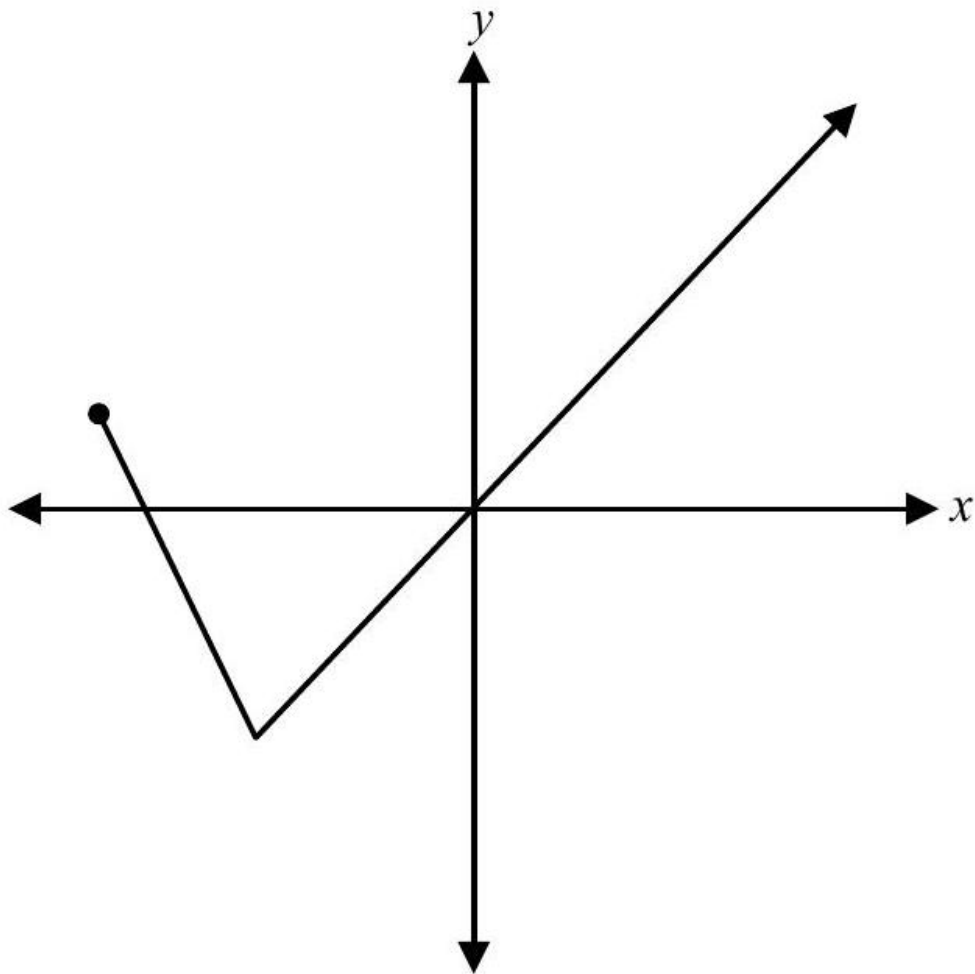
4. (3 points) Solve algebraically:

$$10^{3x} = 7^{x+5}$$

Express your answer correct to 3 decimal places.

5. (2 points) A word contains two Ms, two Es, two Ns, and no other repeated letters.
Suppose one of the Ns is replaced by an M.
Will this replacement result in greater or fewer permutations?
Justify your reasoning.
6. (2 points) There is a group of 16 boys and 12 girls. How many ways can a committee of 3 people be formed if there must be at least 2 girls on the committee?
Express your answer as a whole number.
7. (1 point) sketch the graph of $y = |f(x)|$.





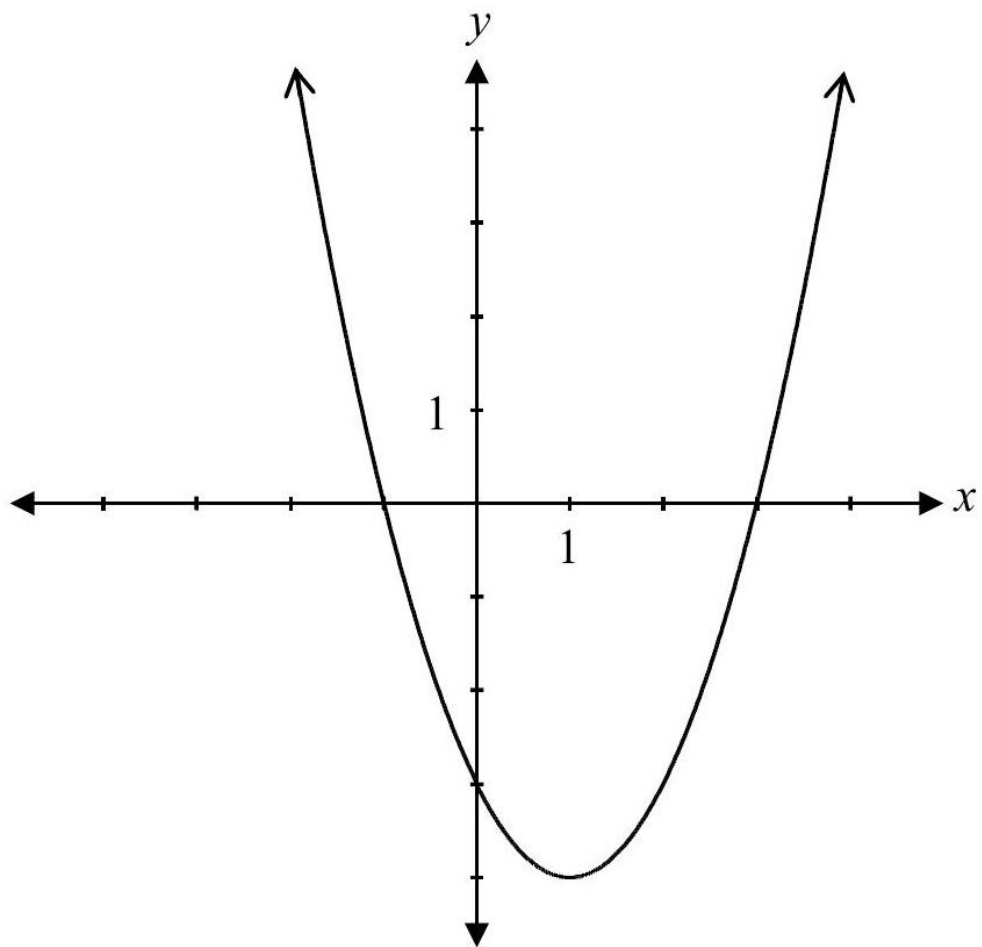
8. (1 point) Claire correctly solves the following equation:

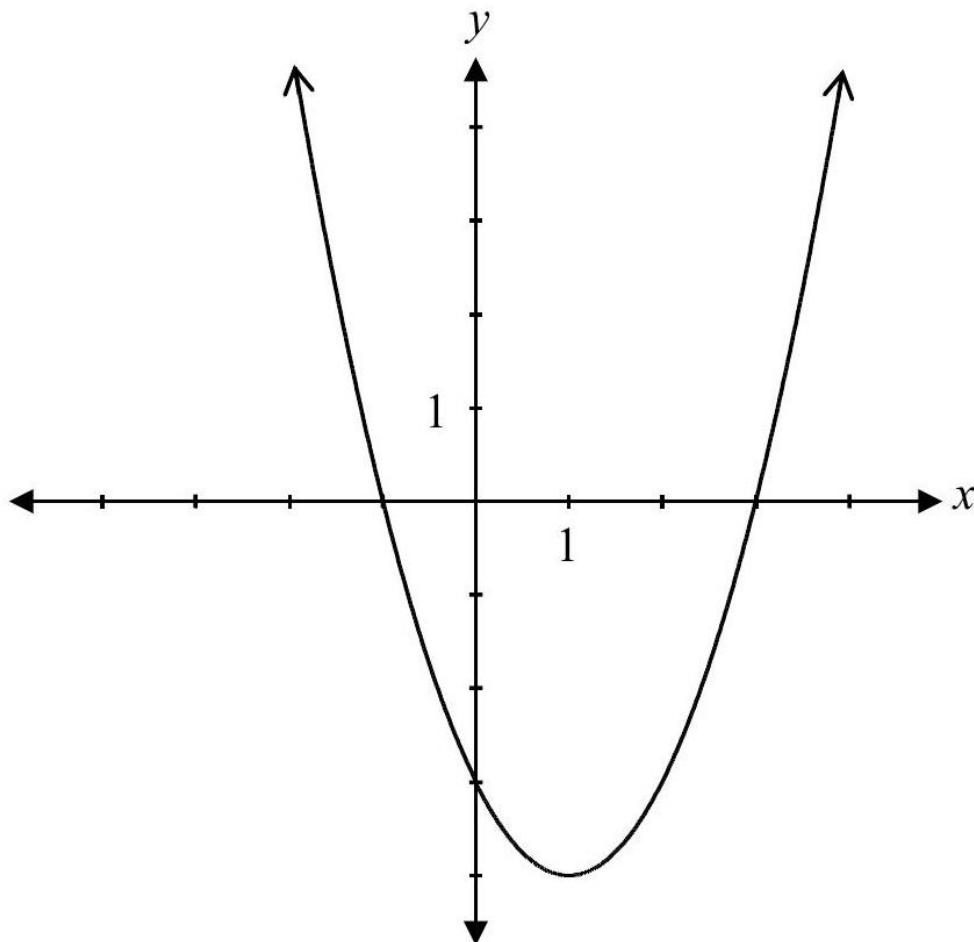
$$\log_2(6 - x) + \log_2(3 - x) = 2$$

She finds two possible values of x : $x = 2$ and $x = 7$.

Identify which one of these values is unacceptable and explain why.

9. (1 point) Given the graph of the function $f(x)$ below, state the domain of $y = \sqrt{f(x)}$.

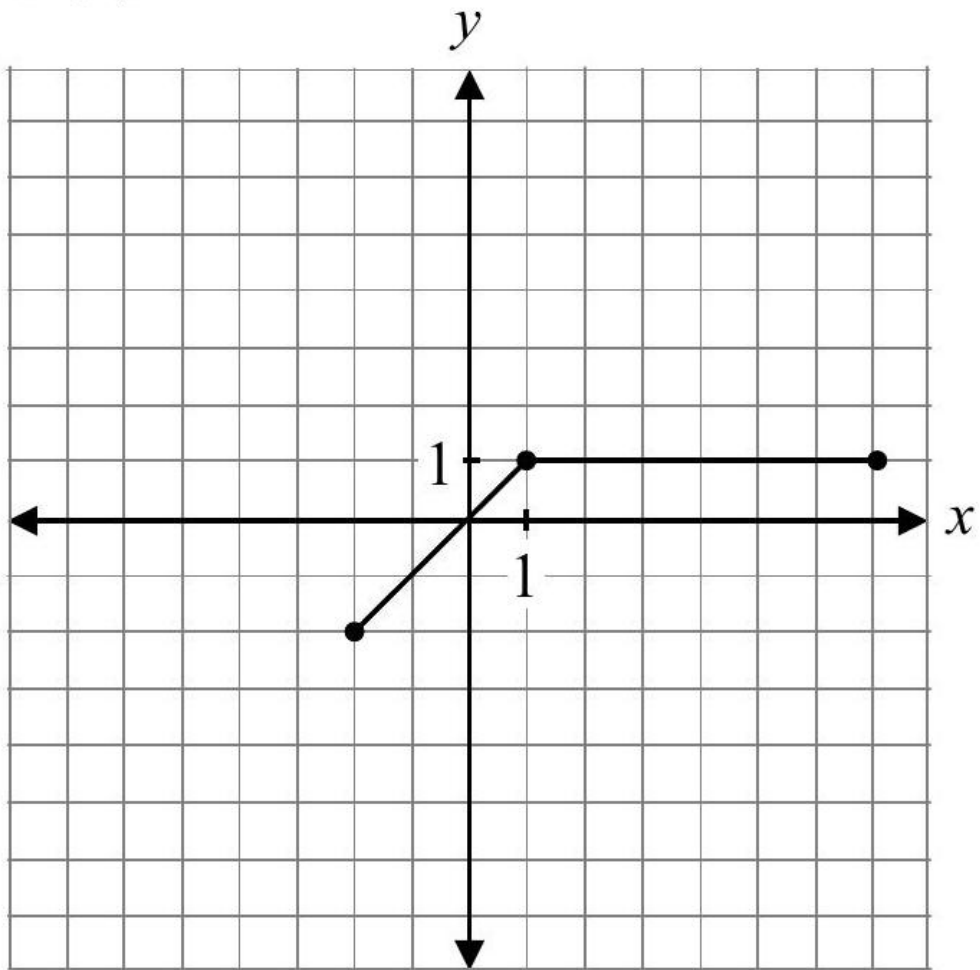


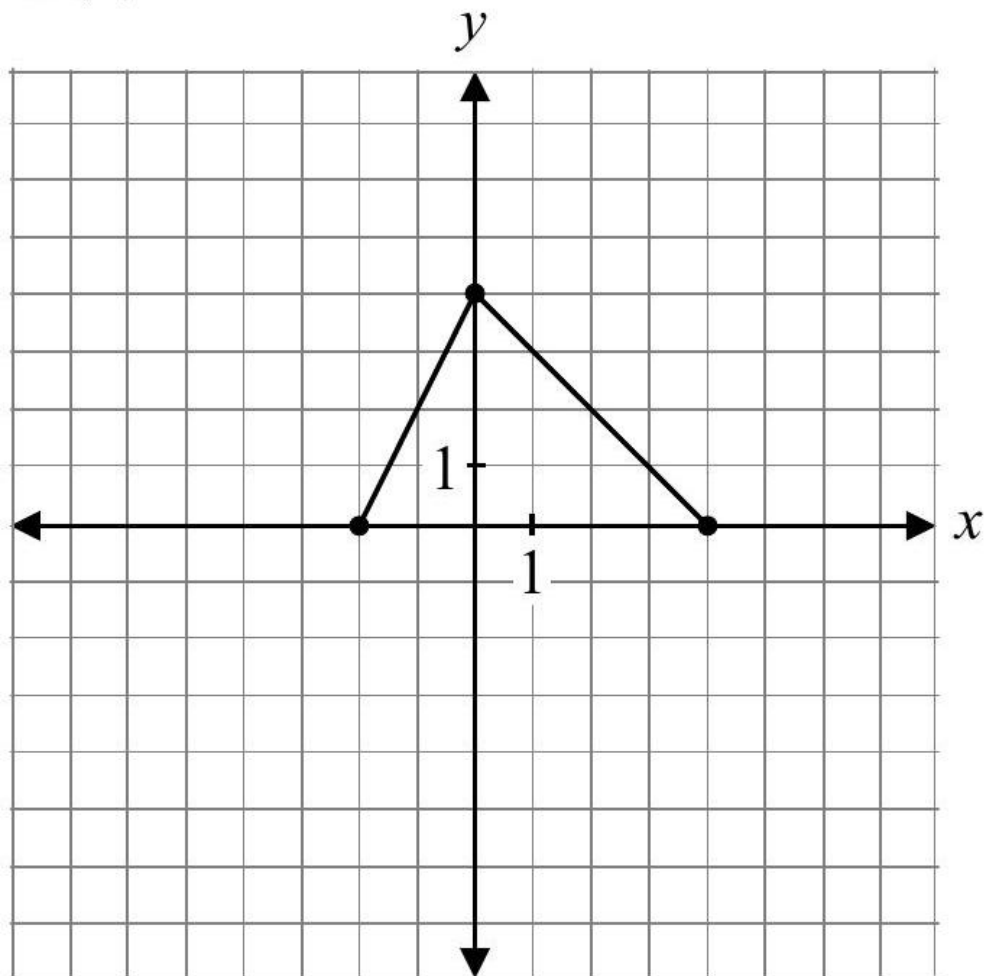


10. (1 point) A school offers 4 different Science courses, 3 different Mathematics courses, and 2 different English courses.
- Julie must select 1 Science course, 1 Mathematics course, and 1 English course. She thinks this creates 9 options for her timetable.
- Show why Julie is incorrect.
11. (1 point) Explain how Pascal's triangle can be used to determine the coefficients in the binomial expansion of $(x + y)^n$.
12. (3 points) Prove the identity below for all permissible values of x :

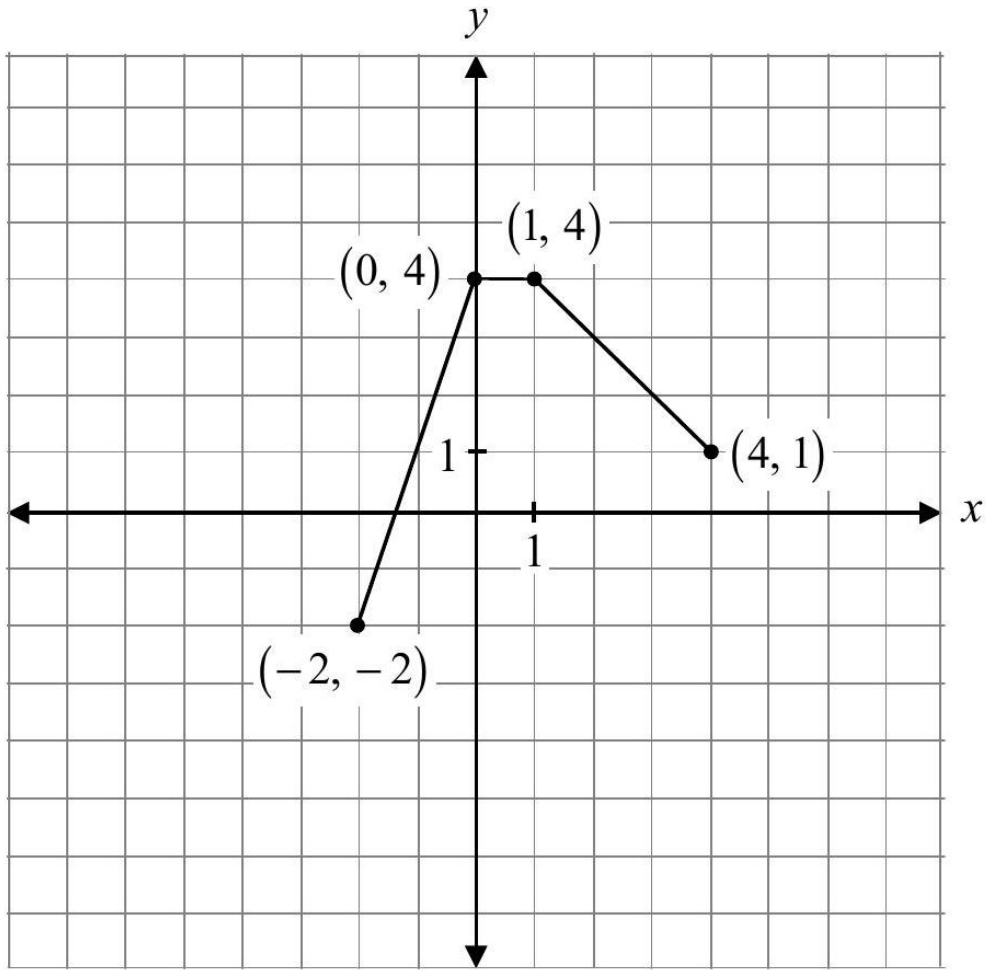
$$\frac{1 + \cos 2x}{\sin 2x} = \cot x$$

13. (2 points) Your classmate, Leo, was absent for one of his math lessons.
- Explain to Leo how to determine the cosecant ratio for an angle in standard position given that $P(-3, -4)$ is a point on the terminal arm of the angle.

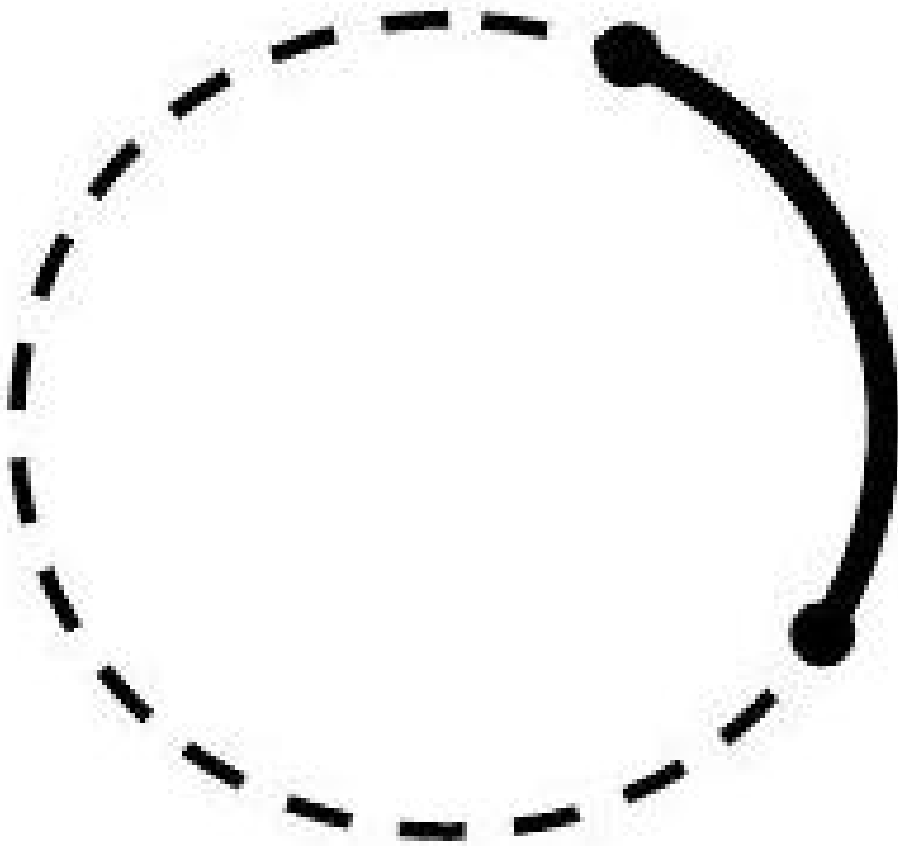


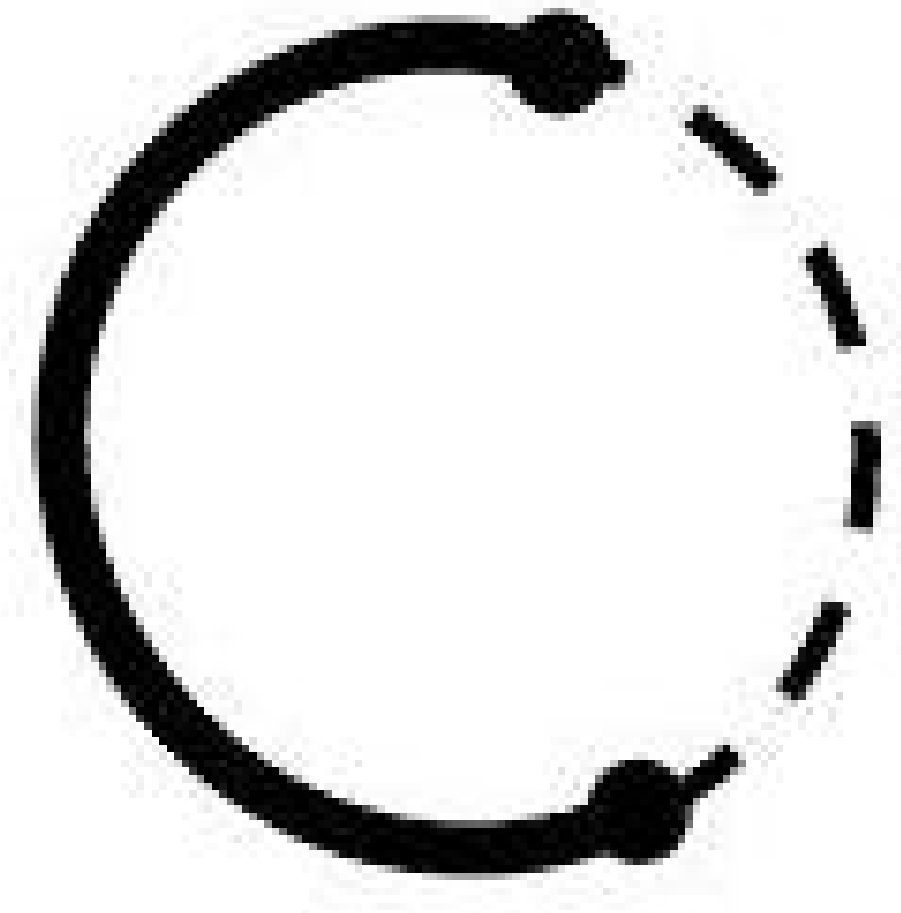


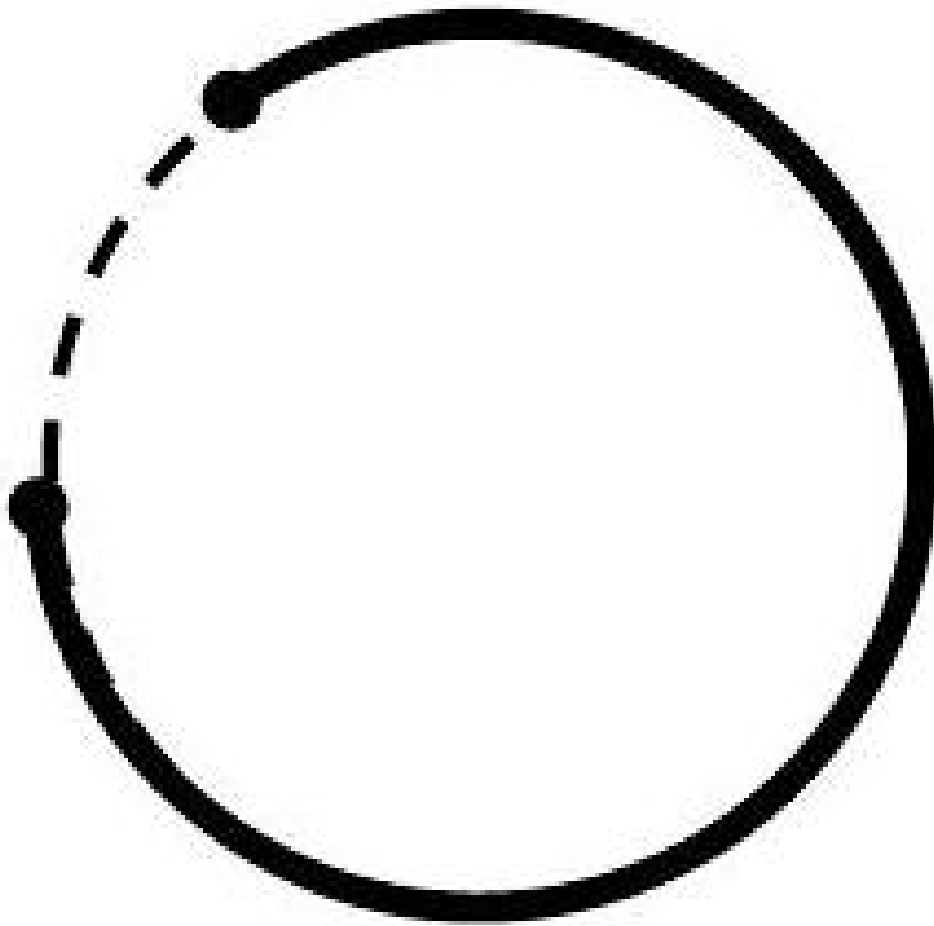
14. (3 points) Sketch the graph of $f(x) + g(x)$.

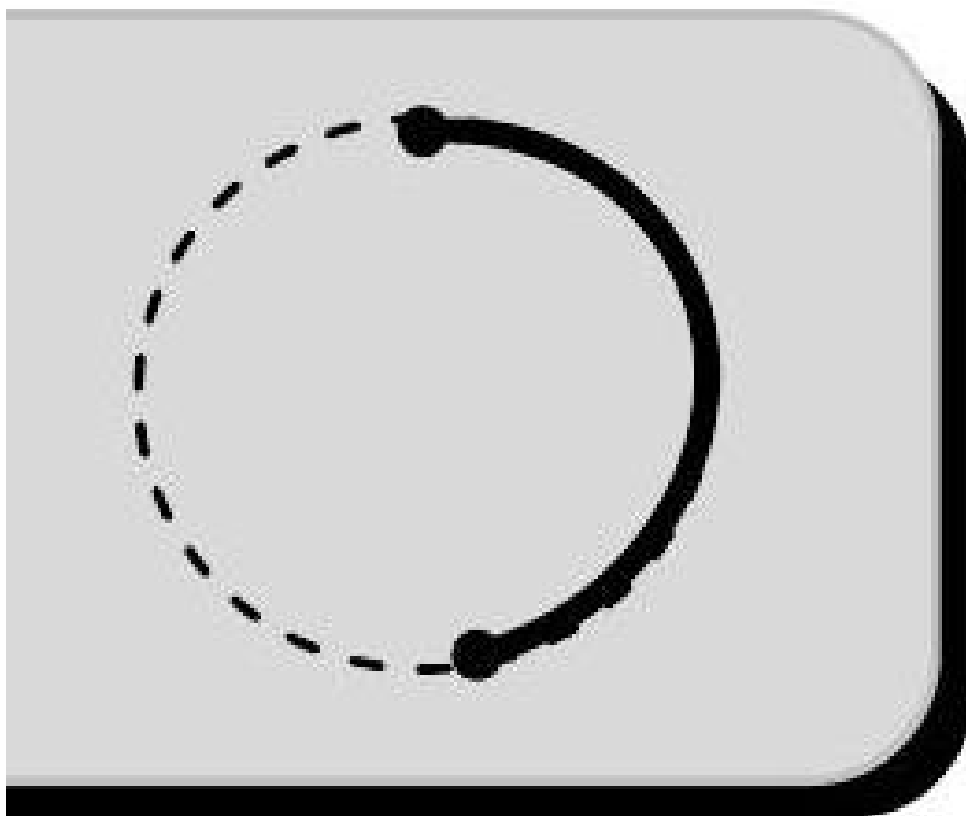


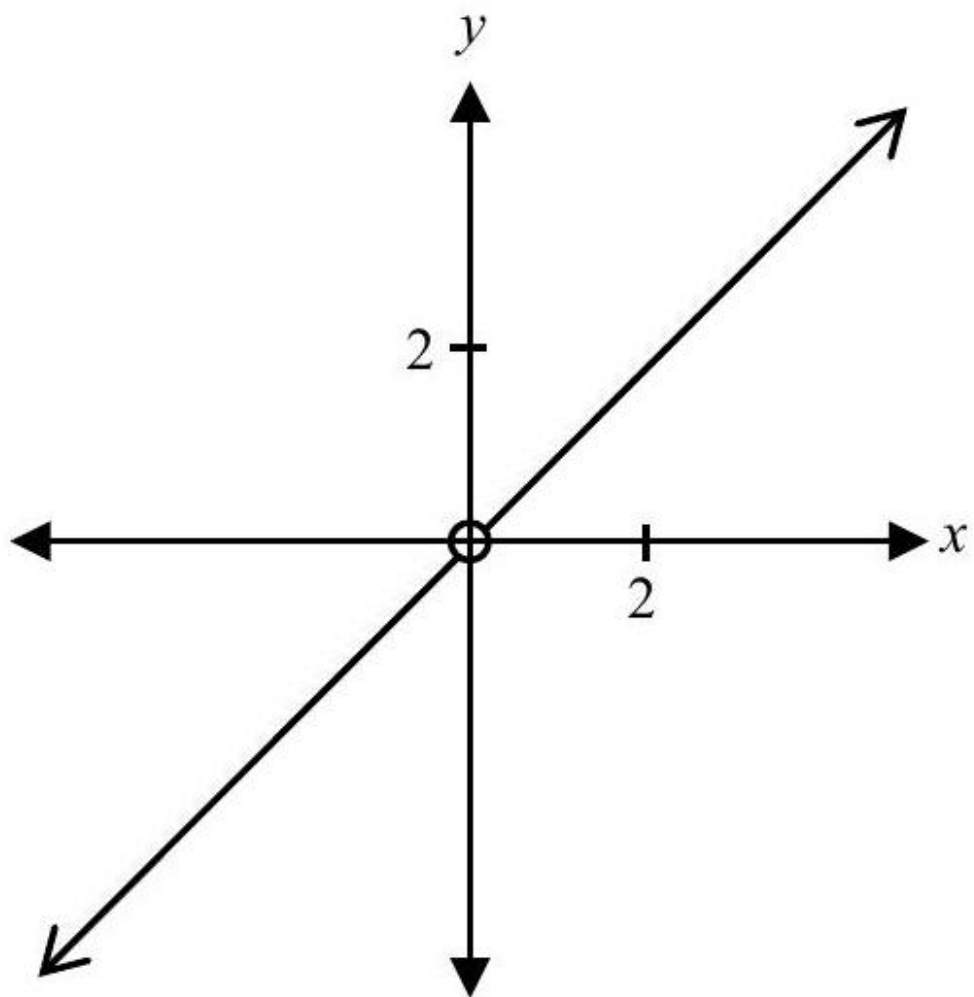
2 marks

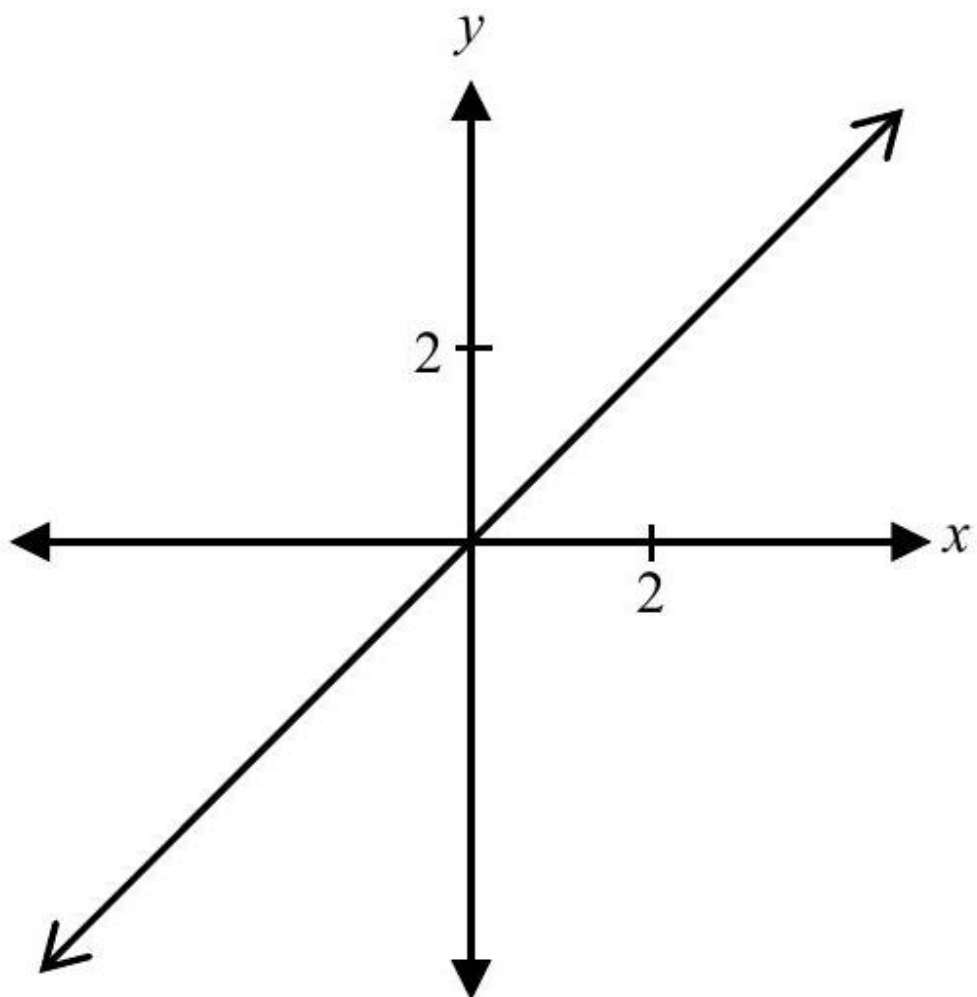


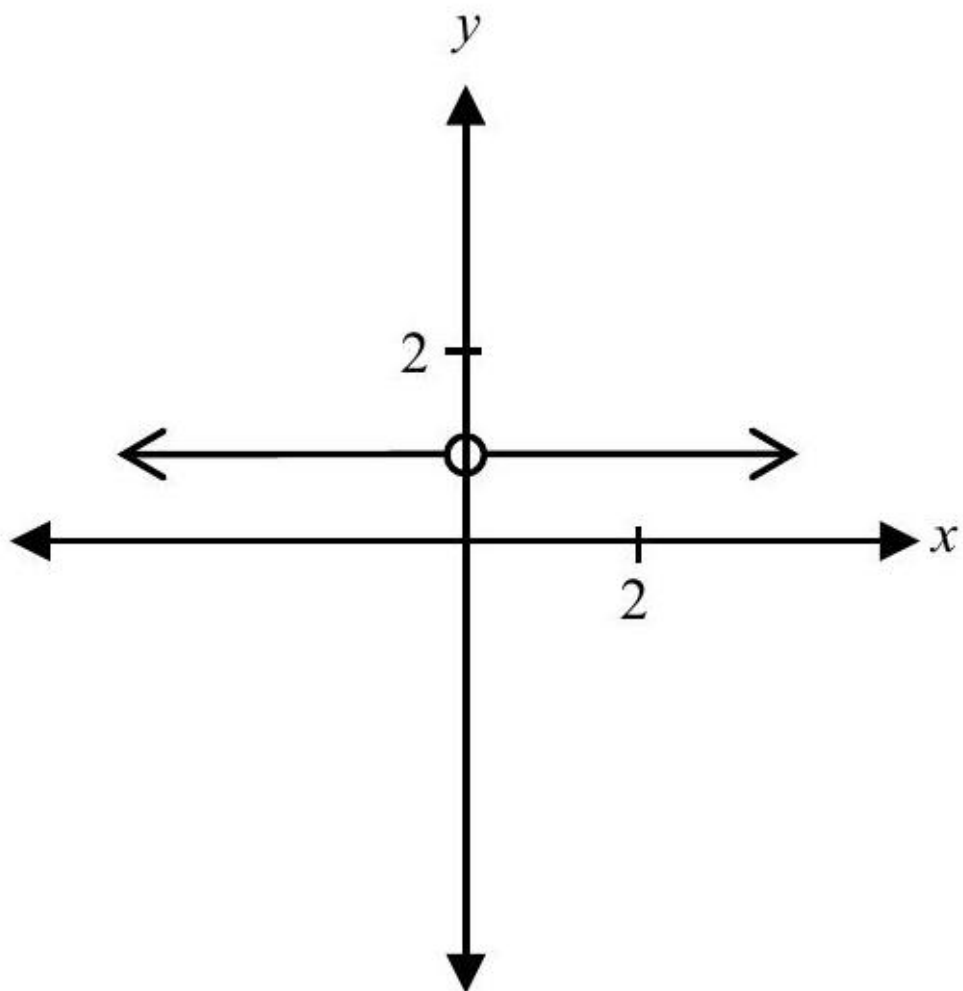


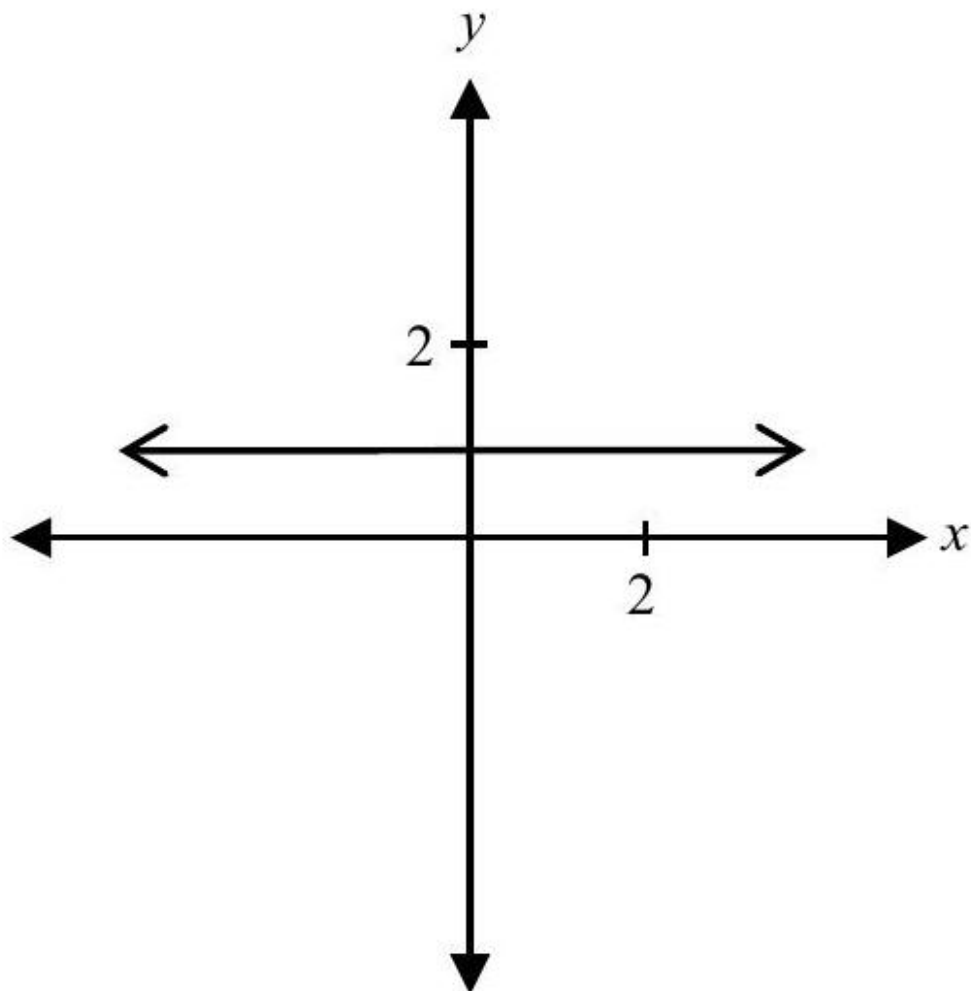












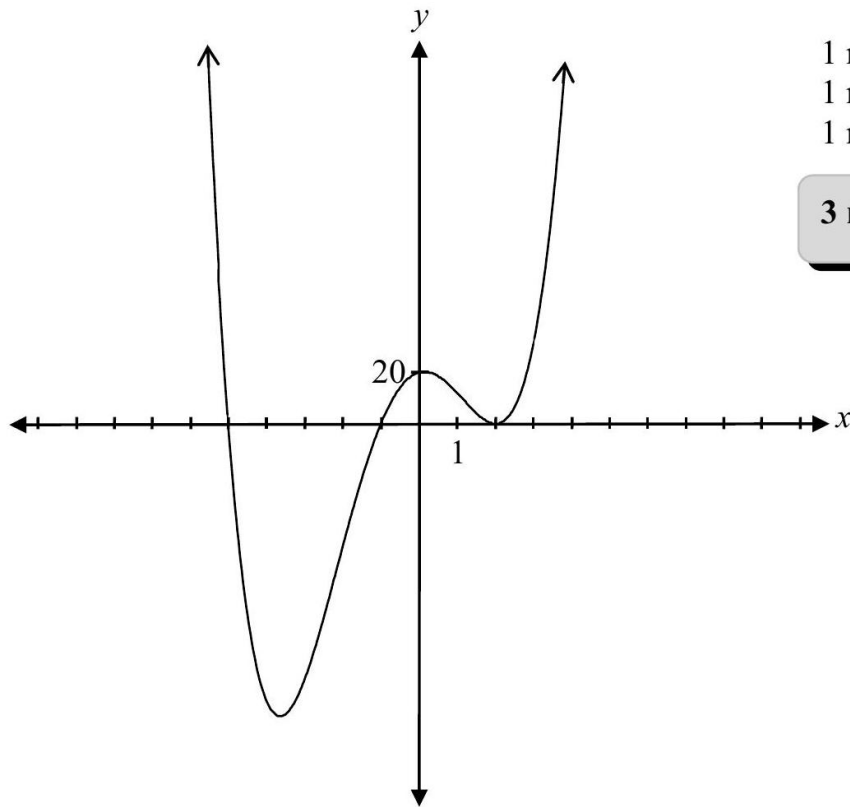
15. (1 point) Identify the value of the x -intercept of the function $y = \ln(x - 2)$.

- a) -1
- b) 0
- c) 2
- d) 3

Given $\log_b a = 3$, give one example of possible values for a and b that make this equation true.

16. (3 points) Sketch the graph of $y = (x + 1)(x - 2)^2(x + 5)$.

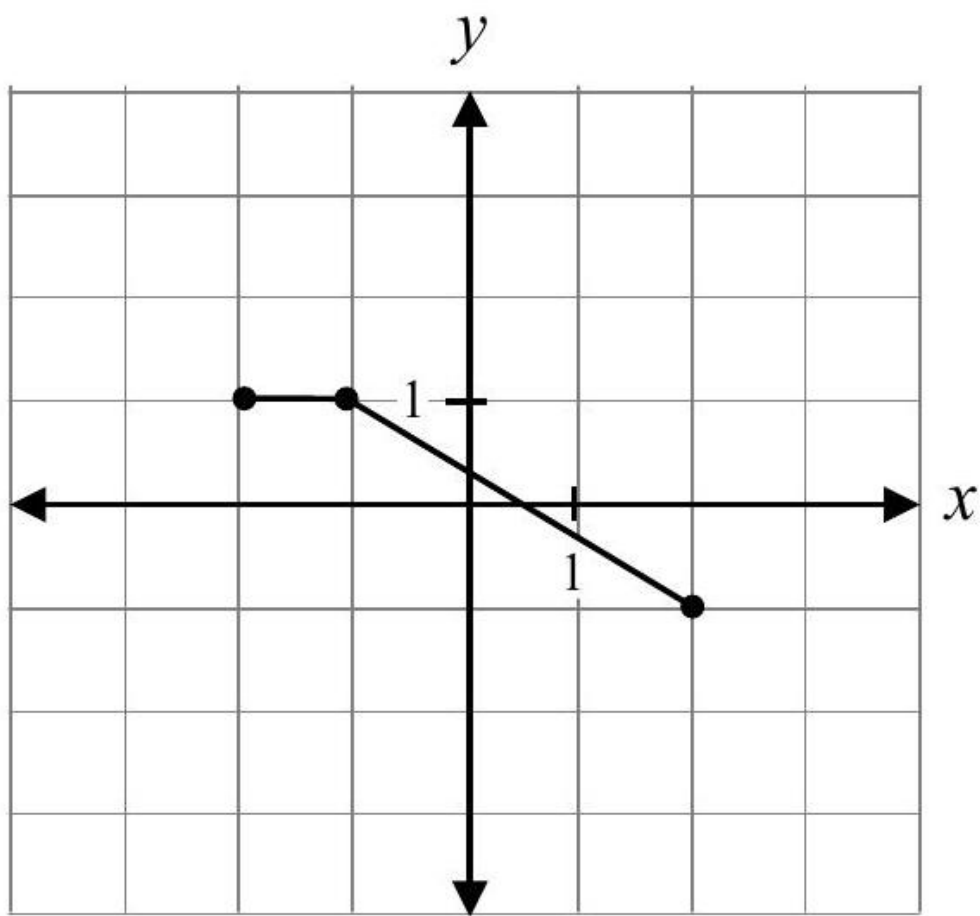
Identify the x -intercepts and y -intercept.

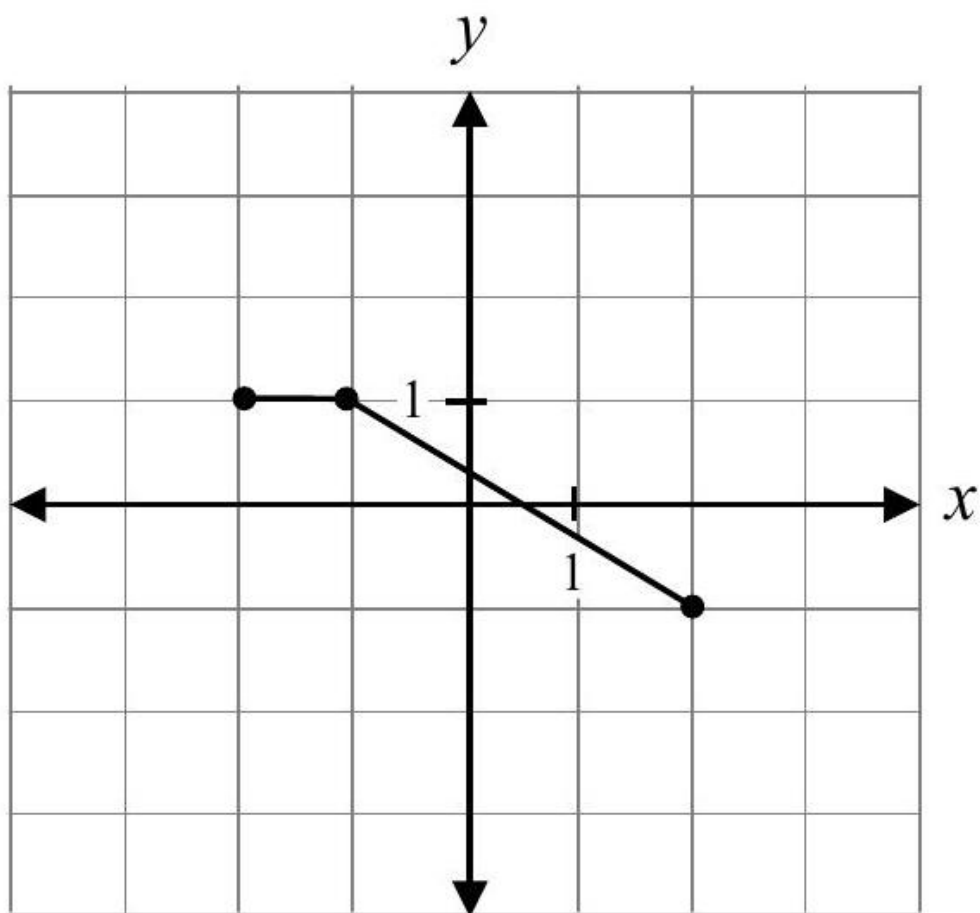


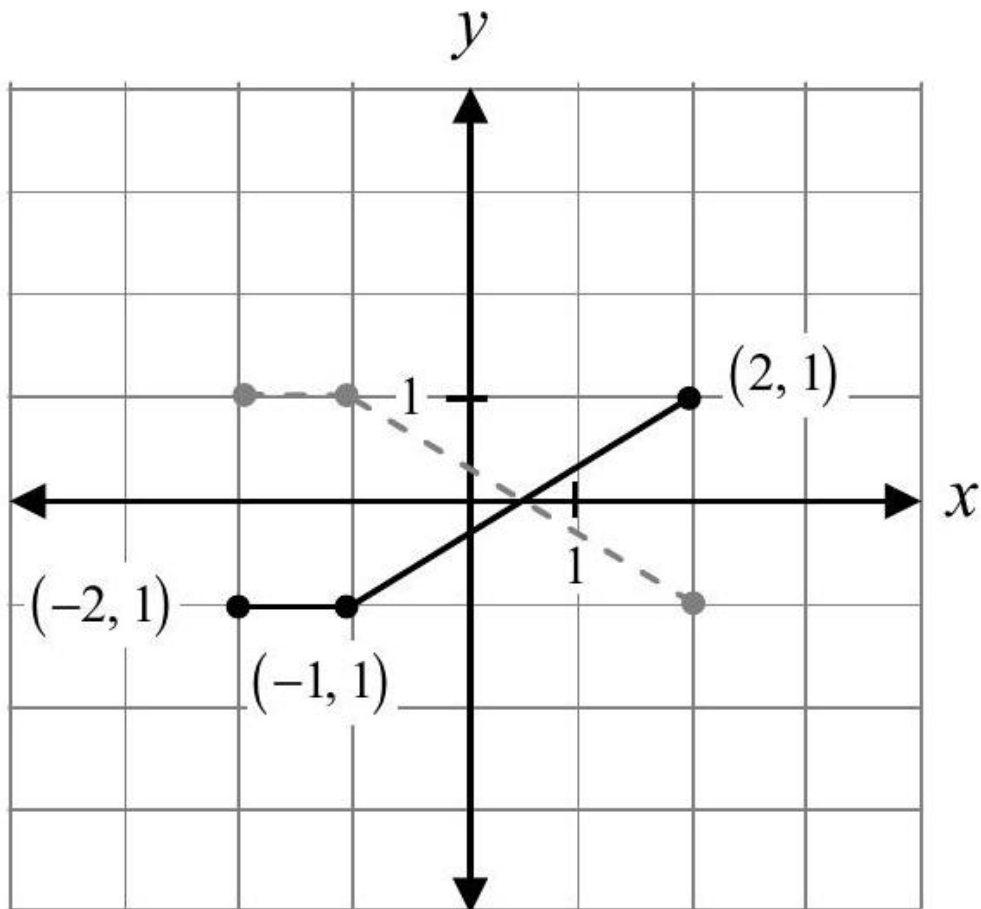
1 mark for x -
1 mark for y -
1 mark for m

3 marks

17. (3 points) sketch the graph of $y = -f(x)$.



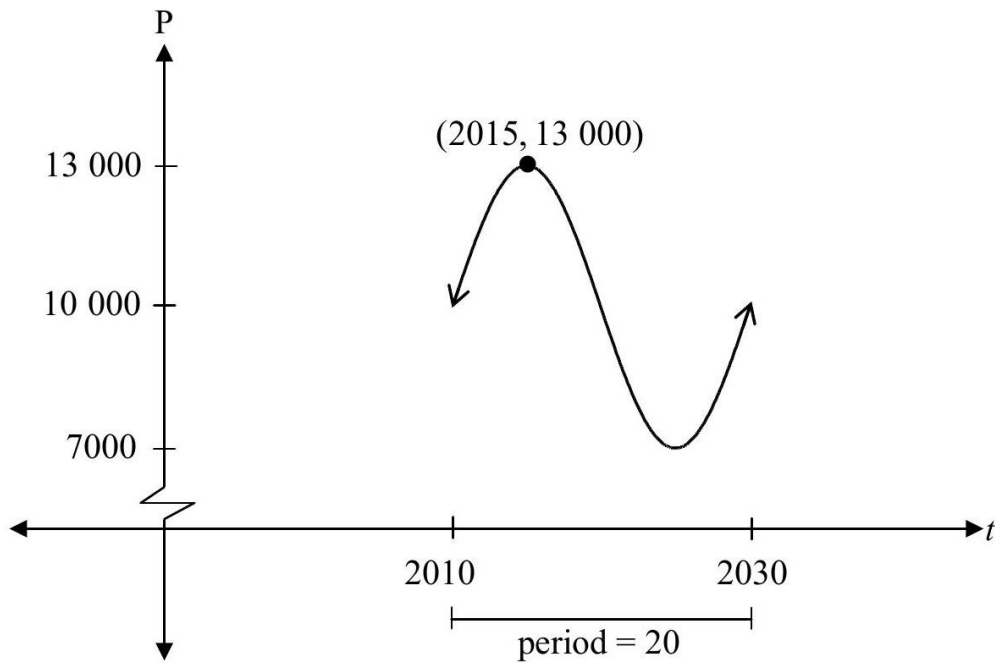




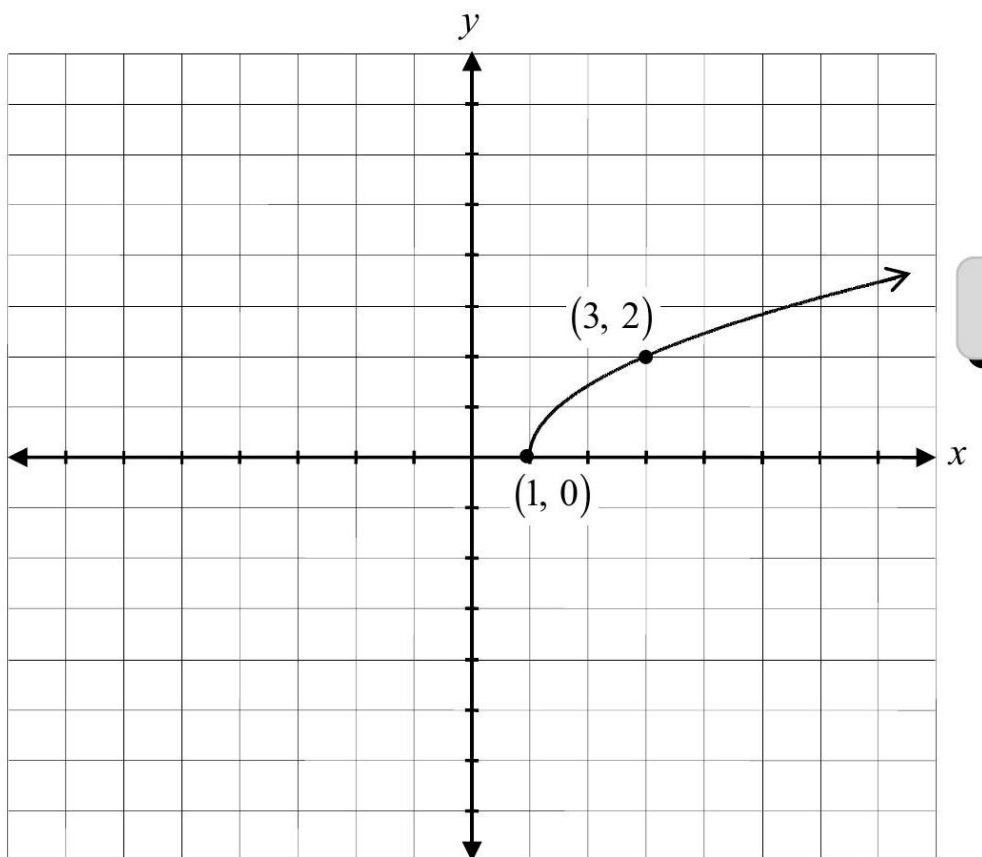
18. (3 points) Given the following sinusoidal equation:

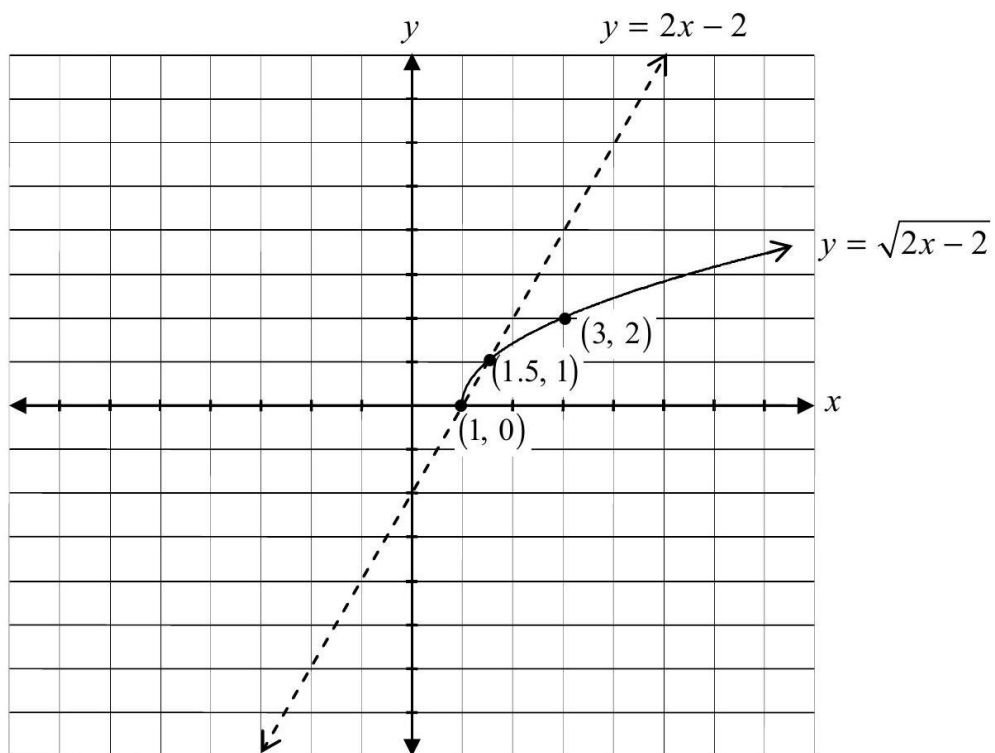
$$P(t) = 3000 \sin \left[\frac{\pi}{10}(t - 2010) \right] + 10000$$

Determine the maximum value of $P(t)$ and a value of t at which this maximum occurs.



19. (1 point) Sketch the graph of $y = \sqrt{2x - 2}$.



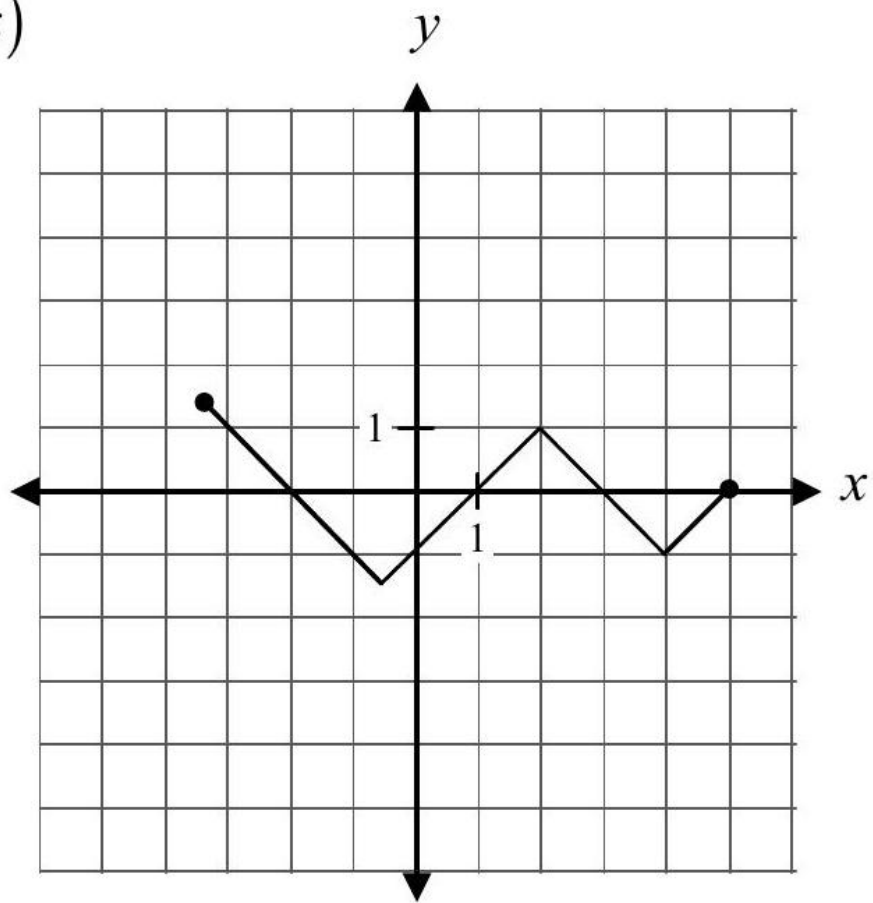


20. (1 point) Determine all non-permissible values of θ over the interval $[0, 2\pi]$.

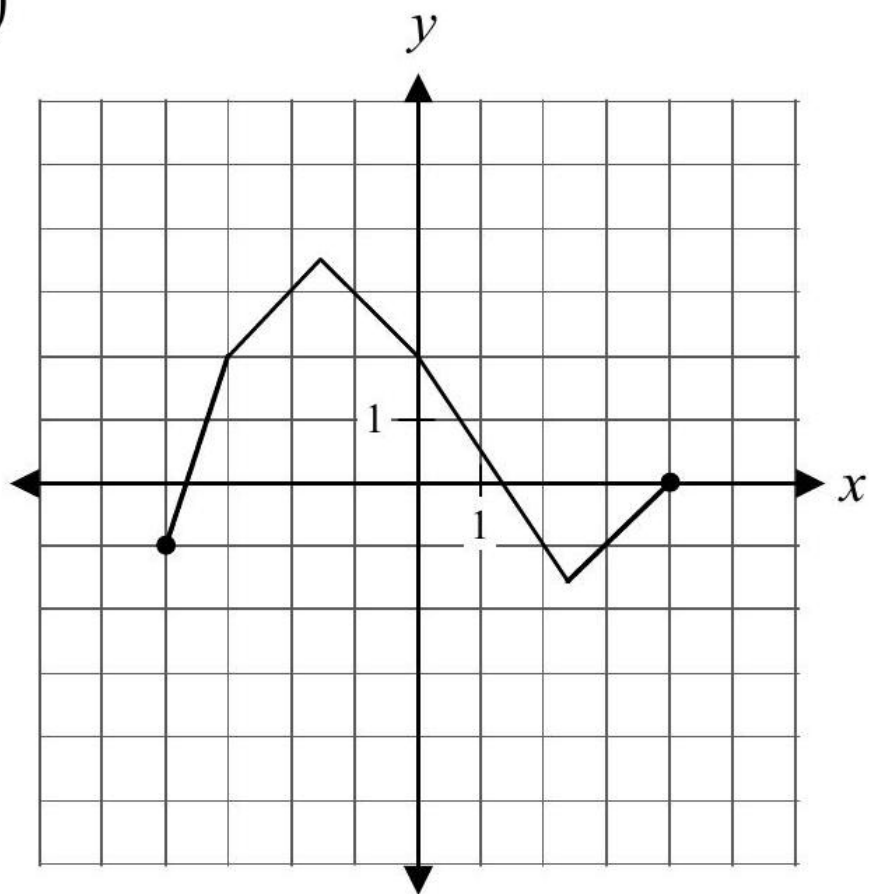
$$\frac{\sin \theta}{1 + \cos \theta} + \csc \theta + \cot \theta$$

Explain your reasoning.

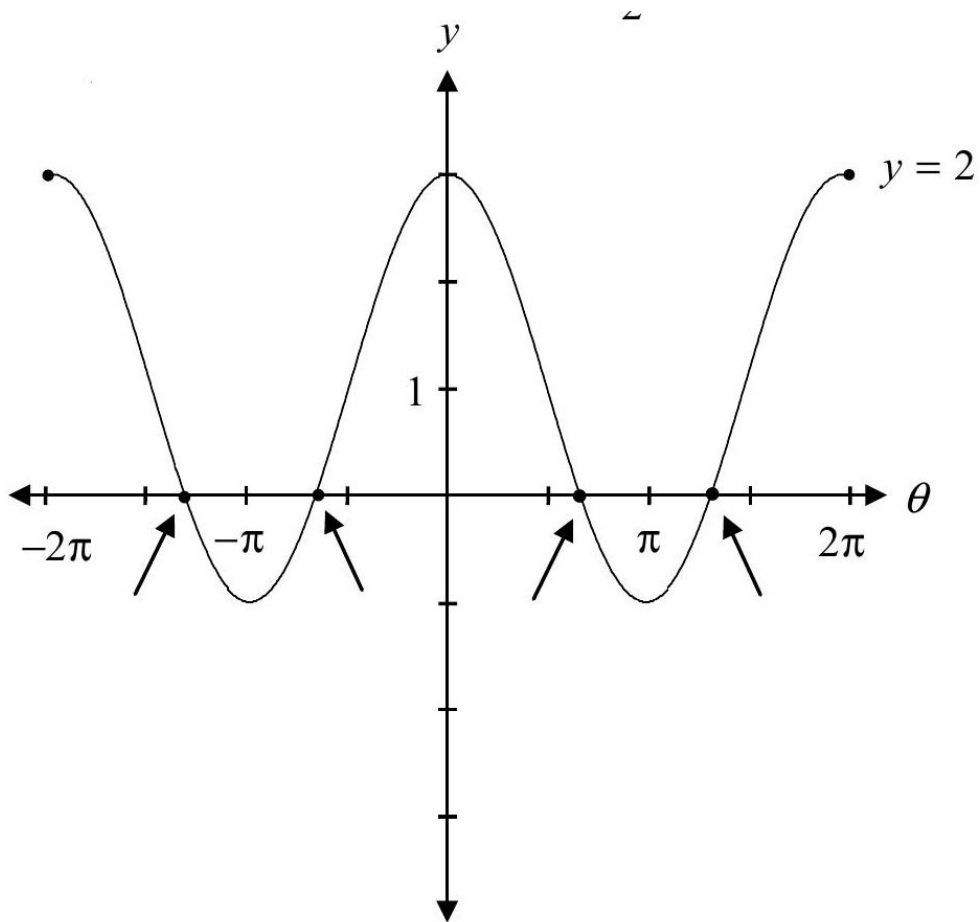
$f(x)$

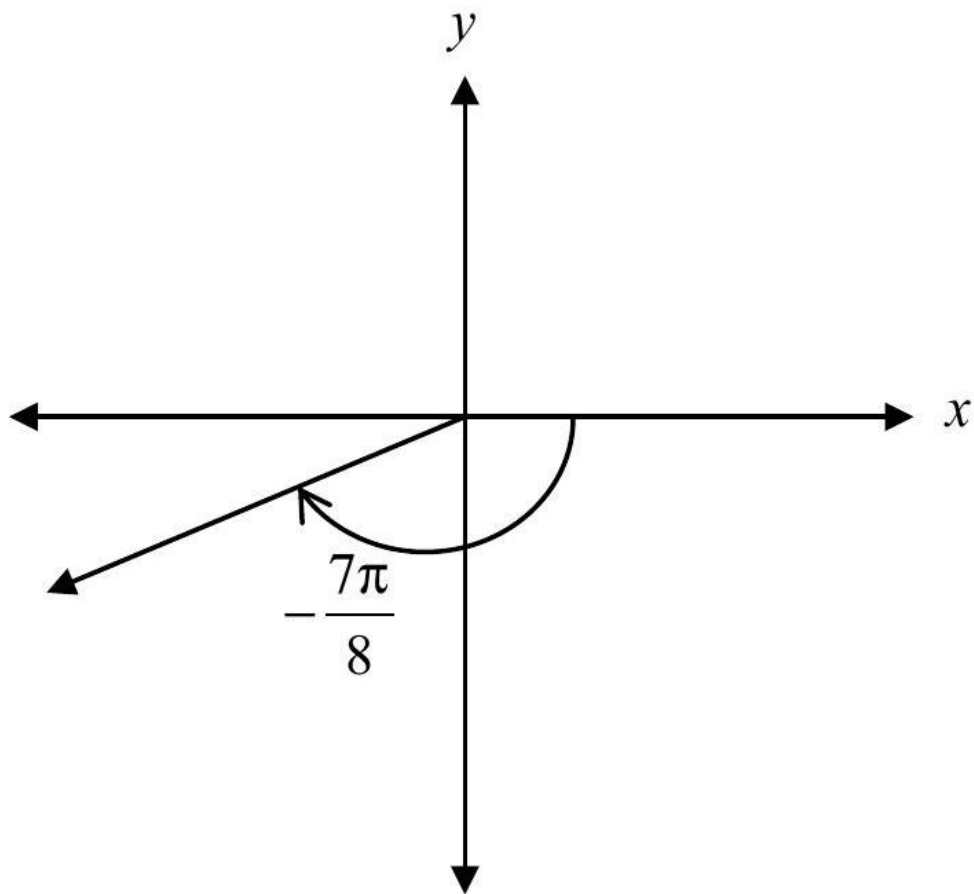


$g(x)$



21. (1 point) Given that $h(x) = 2x^2 + 5x - 3$ and that $h(x) = f(x) \cdot g(x)$, determine $f(x)$ and $g(x)$.

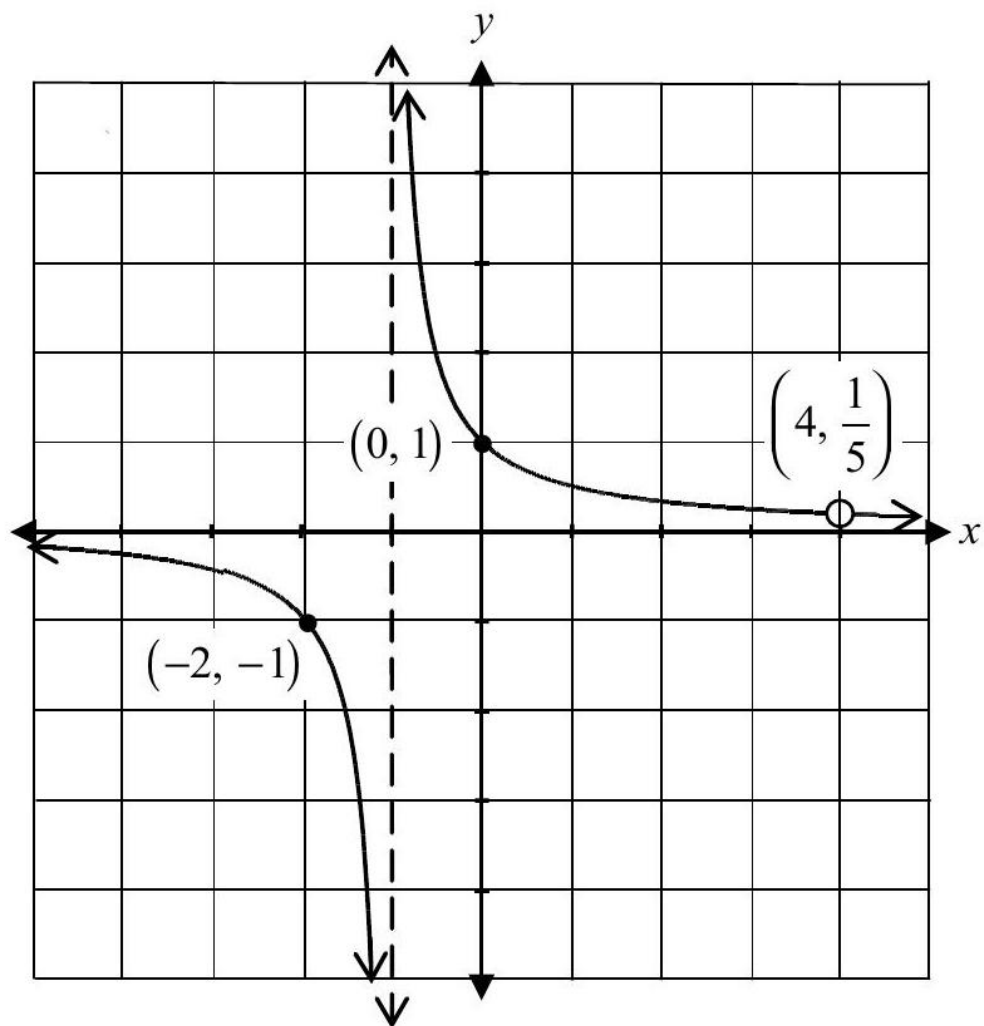




22. (3 points) Determine the exact value of:

$$4 \cos \left(\frac{11\pi}{12} \right)$$

23. (3 points) Sketch the graph of $f(x) = \frac{x-4}{x^2-3x-4}$.



$$\begin{array}{r}
 -2 \overline{) 1 \quad 0 \quad 0 \quad 0 \quad -12 \quad 1} \\
 \underline{\downarrow -2 \quad 4 \quad -8 \quad 16 \quad -8} \\
 1 \quad -2 \quad 4 \quad -8 \quad 4 \quad -7
 \end{array}$$